Lab Sheet-01
BTCS104T
Advanced Python Lab
B.Tech. II Semester (Session 2025-2026)

Objective: To understand the basic concepts of variable, decision and loop control statements etc.

Requirements:

- Text editor or IDE
- Python 3.xx

Assignments:

1. Write a Python program to compare a student's marks with the passing marks and display the result using comparison operators.

```
Marks=int(input("Enter the marks"))
Passing_marks=33
if(Marks>=Passing_marks):
    print("Pass")
else:
    print("Fail")
```

2.Write a program to compare an employee's salary with the company minimum salary and print the comparison results.

```
  employ_salary=int(input("Enter salary"))
minimum_salary=20000
if(employ_salary>minimum_salary):
    print("Salary is greater than minimum salary ",employ_salary)
elif(employ_salary==minimum_salary):
    print("salary is equal to minimum salary")
else:
    print("Salary is less than minimum salary ",employ_salary)
```

3.Write a program to check whether a person's age is equal to, greater than, or less than the voting age.

```
age=int(input("Enter your age:"))
voting_age=18
if(age>voting_age):
    print(f"{age} is greater then voting age")
elif(age<voting_age):
    print(f"{age} is less than voting age")
else:
    print(f"{age} is equal to voting age")
```

4.Write a Python program to compare the price of two products and determine which one is costlier or if both have the same price.

```
product_1=int(input("Enter price of product 1"))
product_2=int(input("Enter price of product 2"))
if(product_1==product_2):
    print("Both product have same prices")
elif(product_1>product_2):
    print("Product 1 is costlier than product 2")
else:
    print("Product 2 is coostlier than product 1")
```

5.Write a program to check whether a customer's account balance is sufficient for a withdrawal amount.

```
Withdrawal_amount=int(input("Enter the withdrawal amount"))
Account_balance=10000
if(Withdrawal_amount<=Account_balance):
    print("Withdrawal successful")
else:
    print("Insufficient Balance")
```

6.Write a program to compare a student's attendance percentage with the required attendance for exam eligibility.

```
Attendace=int(input("Enter your attendance percentage"))
minimum_attendance=75
if(Attendace>=minimum_attendance):
    print("Eligible for Exam")
else:
    print("Not Eligible for Exam")
```

7.Write a Python program to compare current temperature with a normal temperature range.

```
# Normal temperature range
normal_min = 18
normal_max = 30

# Take current temperature from user
current_temp = float(input("Enter current temperature: "))

# Compare temperature with normal range
if current_temp < normal_min:
    print("Temperature is below normal range ❄️")
elif current_temp > normal_max:
    print("Temperature is above normal range 🔥")
else:
    print("Temperature is within normal range ✅")
```

8.Write a program to check whether an employee's working hours are equal to, more than, or less than the required working hours.

```
Working_hour=int(input("Enter your working hours"))
Minimum_working_hour=8
if(Working_hour>Minimum_working_hour):
    print("Working hour is more than minimum working hour")
elif(Working_hour<Minimum_working_hour):
    print("Working hour is less than minimum working hour")
else:
    print("Working hour equal to minimum working hour")
```

9.Write a program to compare a person's monthly income with the tax exemption limit.

```
tax_limit=25000
income=int(input("Enter your Income :"))
if(income<=tax_limit):
    print("No tax")
else:
    print("Tax applicable ")
```

10.Write a Python program to compare two delivery times and determine which delivery is faster or if both are the same.

```
Delievery_1=int(input("Enter delivery 1 time in minutes:"))
Delievery_2=int(input("Enter delivery 2 time in minutes:"))
if(Delievery_1<Delievery_2):
    print("Delievery 1 is faster than Delievery 2")
else:
     print("Delievery 2 is faster than Delievery 1")
```

11.Write a Python program that takes a number from the user and checks whether the number is positive and even using logical AND. The program should repeat using a while loop until the user enters NO.

```
while True:
    number=int(input("Enter number :"))

    if(number>=0 and number%2==0):
        print("Number is positive and even")
    else:
        print("Number is negative and  not even")
    choice=input("Do you want to continue (YES/NO)").upper()
    if(choice=="NO"):
        print("Exiting the loop")
        break
```

12.Write a Python program that accepts marks of five subjects using a for loop and checks whether the student has passed all subjects (marks ≥ 40) using logical AND. Stop taking input using break if any subject is failed.

```
all_passed=True
for i in range(1,6):
    marks=int(input(f"Enter subject{i} marks:"))
    if marks <40:
        print("Stundent has failed")
        all_passed=False
        break
if all_passed:
    print("Student has passed all subjects")
```

13.Write a menu-driven Python program that takes age and citizenship status as input and checks voting eligibility using logical AND. The menu should repeat using a while loop until the user chooses Exit.

```
while True:
    print("/nVoting Eligibility Check")
    print("1.Check Voting Eligibility")
    print("2.Exit")
    choice=input("Enter your choice (1/2): ")
    if choice=="1":
        age=int(input("Enter your age: "))
        citizenship=input("Enter your citizenship status").lower()
        minimum_age=18
        citizenship_required="indian"
        if(age>=minimum_age and citizenship_required==citizenship):
            print("Eligible for voting")
        else:
            print("Not eligible for voting")
    elif choice=="2":
        print("Exiting the program")
        break
```

14.Write a Python program that takes salary and experience as user input and checks bonus eligibility using logical AND. Display the result using a ternary operator.

```
salary=int(input("Enter your salary:"))
Experience=int(input("Enter how many year experience you have:"))
bonus=("Bonus Eligible"if salary>=30000 and Experience>=3 else "Not Eligible
for Bonus")
print(bonus)
```

15.Write a Python program that takes a year as input and checks whether it is a leap year using logical AND and OR operators.

```
year=int(input("Enter year:"))
if((year%4==0 and year%100!=0)or (year%400==0)):
    print("Leap year")
else:
    print("Not leap year")
```

16.Write a Python program that takes a number from the user and checks whether it is divisible by 3 or 5 using logical OR. Allow multiple checks using a while loop.

```
while True:
    num=int(input("Enter a number:"))
    if(num%3==0 or num%5==0):
        print("Number is divisible by 3 and 5")
    else:
        print("Number is not divisible by 3 and 5")
    choice=input("Do you want to check another number(YES?NO)").upper()
    if choice=="NO":
        break
```

17.Write a Python program that accepts state-level and national-level player status and checks eligibility for sports quota using logical OR.

```
state=input("Are you state level player(YES/NO):").lower()
national=input("Are you national level player(YES/NO):").lower()
if state=="yes" or national=="yes":
    print("Eligible for sport Quota")
else:
    print("Not Eligible for sport Quota")
```

18.Write a menu-driven Python program that takes purchase amount and membership status as input and checks discount eligibility using logical OR. The program should repeat until the user selects Exit.

```
while True:
    print("\nDiscount Eligiblity checking")
    print("1.checking")
    print("2.Exit")
    choice=int(input("ENTER YOUR CHOICE(1/2)"))
    if choice==1:
        amount=int(input("Enter your purchase Amount:"))
        Membership_status=input("Enter your membership
status(Yes/no)").lower()
        if(amount>=5000 or Membership_status=="yes"):
            print("Eligible for Discount")
        else:
```

```
            print("Not Eligible for Discount")
    elif choice==2:
        print("Exiting")
        break
```

19.Write a Python program that takes degree and experience as input and checks job eligibility using logical OR. Display the result using a ternary operator.

```
degree=input("Enter your degree in cs:").upper()
experience=int(input("Enter your experience:"))
eligiblity="Eligible"if degree=="CS" or experience>=3 else "Not eligible"
print(eligiblity)
```

20.Write a Python program that accepts a day name from the user and checks whether it is a weekend using logical OR.

```
day=input("Enter day :").lower()
if(day=="sunday" or day=="saturday"):
    print("Weekend")
else:
    print("Not weekend")
```

21.Write a Python program that takes age as input and checks whether the person is not eligible to donate blood using the NOT operator.

```
age=int(input("Enter your age :"))
if not(age>=18):
    print("Not Eligible for donating blood")
else:
    print("Eligible for donating blood ")
```

22.Write a Python program that takes login status as input and reverses the login condition using the NOT operator.

```
login=input("Are you logged in?(yes/no):").lower()
if not(login=="yes"):
    print("User is not logged in")
else:
    print("User is logged in")
```

23.Write a Python program that takes a number as input and checks whether the number is not positive using the NOT operator with comparison operators.

```
num=int(input("Enter a number:"))
if not(num>=0):
    print("Not positive")
else:
    print("Positive")
```

24.Write a Python program that accepts a user role and checks whether the user is not an admin using the NOT operator.

```
role=input("Enter your role:").lower()
if not(role=="admin"):
    print("Not admin")
else:
    print("Admin")
```

25.Write a Python program that takes attendance status as input and checks whether the student is not absent using the NOT operator.

```
status=input("Enter attendance status(present/absent): ").lower()
if not(status=="absent"):
    print("Student is not absent")
else:
    print("Student is absent")
```

26.Write a Python program that takes income, credit score, and defaulter status as input and checks loan eligibility using logical AND and NOT operators.

```
income=int(input("Enter your income:"))
credit_score=int(input("Enter your credit score:"))
Defaulter_status=input("Are you defaulter(Yes/No)").lower()
if income>=25000 and credit_score>=700 and not(Defaulter_status=="yes"):
    print("Loan appproved")
else:
    print("Loan Rejected")
```

27.Write a Python program that accepts login status and blocked status as input and checks system access using logical AND and NOT operators.

```
login_status=input("Are you logged in(yes/no):").lower()
blocked_status=input("Are you blocked (yes/no):").lower()
if login_status=="yes" and not(blocked_status=="yes"):
    print("Access granted")
else:
    print("Access Denied")
```

28.Write a Python program that takes exam result, sports quota status, and detention status as input and checks student promotion using logical AND, OR, and NOT operators.

```
result=input("Enter exam result?(pass/fail):").lower()
sports=input("Do you have sports quota?(yes/no):").lower()
detained=input("Are you detained?(yes/no):")
if((result=="pass" or sports=="yes")and not(detained=="yes")):
```

```
    print("promoted")
else:
    print("Not promoted")
```

29.Write a Python program that accepts permit status, emission test status, and emergency vehicle status as input and checks vehicle entry into a city using logical AND and OR operators.

```
permit_status=input("Do you have permit ? (yes/no)L:").lower()
emission_status=input("Is emmission test passed?(yes/no):").lower()
emergency=input("Is it an Emergency vehicle?(yes/no):").lower()
if (permit_status=="yes" or emission_status=="yes") or emergency=="yes":
    print("Vehicle Allowed to Enter ")
else:
    print("Vehicle Not Allowed")
```

30.Write a Python program that takes senior employee status, medical reason status, and notice period status as input and checks work-from-home eligibility using logical AND, OR, and NOT operators. The program should repeat using a while loop until the user enters NO.

```
while True:
    senior=input("Senior Employ(yes/no):").lower()
    medical=input("Medical Problem(yes/no):").lower()
    notice=input("Are you on notice period(yes/no)").lower()
    if (senior=="yes" or medical=="yes") and notice!="yes":
        print("Work From Home Approved")
    else:
        print("Work From Home Not Approved")
    choice=input("Do you want to continue?(yes/no)").lower()
    if choice=="no":
        print("Exit")
        break
```